# Akshay Pranav - Project Portfolio Page

## Overview

ItemTasker is a CLI-based Stock Keeping Unit (SKU) Ticketing System designed to manage warehouse inventory and track operational tasks across individual SKUs from the terminal.

## Summary of Contributions

### Code contributed

RepoSense link

### Enhancements Implemented

**Edit SKU**

Implemented `SKUCommandHandler#handleEditSku()` and `SKU#setLocation()` to relocate SKUs across grid cells without losing attached tasks (Moderate Difficulty). Uses in-place mutation to preserve all task associations, with null-checks, case-insensitive lookup, and `Location` enum validation.

**Edit Task**

Implemented `TaskCommandHandler#handleEditTask()` and `SKUTaskList#editSKUTask()` for partial task updates, only fields explicitly provided (`d/`, `p/`, `t/`) are modified, preserving the rest (High Difficulty). Backed by a 7-stage validation chain with early-return error handling at each stage.

**Sort Tasks**

Designed and implemented `TaskSorter.java` as a standalone, SRP-compliant class to sort a SKU's task list by `date`, `priority`, or `status` in ascending or descending order (Moderate Difficulty). Operates on a defensive copy and builds a type-safe `Comparator` via a switch, with field validation before instantiation.

**DateValidator**

Implemented `DateValidator.java` as a shared static utility for `YYYY-MM-DD` date validation, reused across `addskutask`, `edittask`, and other handlers (Low Difficulty). Validates via regex then `LocalDate.parse()` to catch invalid strings and impossible dates like `2026-02-30`, with a non-blocking warning for past dates.

**Mark / Unmark Task**

Implemented `TaskCommandHandler#handleMarkTask()` and `handleUnmarkTask()` with duplicate state checks where re-marking a completed task returns an informational message instead of a misleading success message (Moderate Difficulty). Both handlers share the same validation structure, with assertion guards in `SKUTask#mark()` / `unmark()` acting as a secondary safety check.

### Contributions to the User Guide (UG)

Authored a FAQ entry on multi-field task editing via `edittask`, and compiled the Command Summary table covering all 16 commands with formats and examples.

## Contributions to the Developer Guide (DG)

**Architecture**

Wrote the Architecture section describing the overall component structure (UI → Parser → CommandRunner → Handlers → Model → Storage) with the accompanying architecture diagram.

**Edit SKU / Edit Task**

Authored the Edit SKU and Edit Task implementation sections, covering step-by-step scenario walkthroughs, sequence and architecture diagrams, and design considerations such as in-place mutation vs delete-and-recreate.

## Contributions to UML diagrams

`architecture.puml`, `edit-sku-sequence.puml`, `edit-sku-architecture.puml`, `edit-task-sequence.puml`, `edit-task-architecture.puml`, `markTaskSequence.puml`, `mark-unmark-architecture.puml`, `unmarkTaskSequence.puml`, `sort-task-sequence.puml`, `sort-task-architecture.puml`

## Contributions to team-based tasks

1. **CommandRunner Refactor:** Split the monolithic `CommandRunner` into `TaskCommandHandler`, `SKUCommandHandler`, and `ViewCommandHandler`, applying SRP to improve maintainability for the whole team.
2. **UML Diagram Standardisation:** Overhauled all PlantUML diagrams across the DG by introducing a consistent `style.puml` base, fixing formatting inconsistencies and improving overall readability.
3. **Team Coordination:** Facilitated discussions on architecture and feature design, identified bugs in teammates' code, and assisted in resolving them across the codebase.

## Contributions Beyond the Project Team

**Peer Quality Assurance (PE-D):** Tested a peer team's application during the Practical Exam Dry Run, ranking in the top 10% of bug reporters by identifying critical functionality flaws and documentation bugs.

# Contributions to the User Guide (Extracts)

## FAQ

**Q**: Can I edit multiple fields of a task at once? **A**: Yes. The `edittask` command accepts any combination of `d/` (due date), `p/` (priority), and `t/` (description) flags simultaneously. Only the fields you specify will be updated. All other fields are preserved.

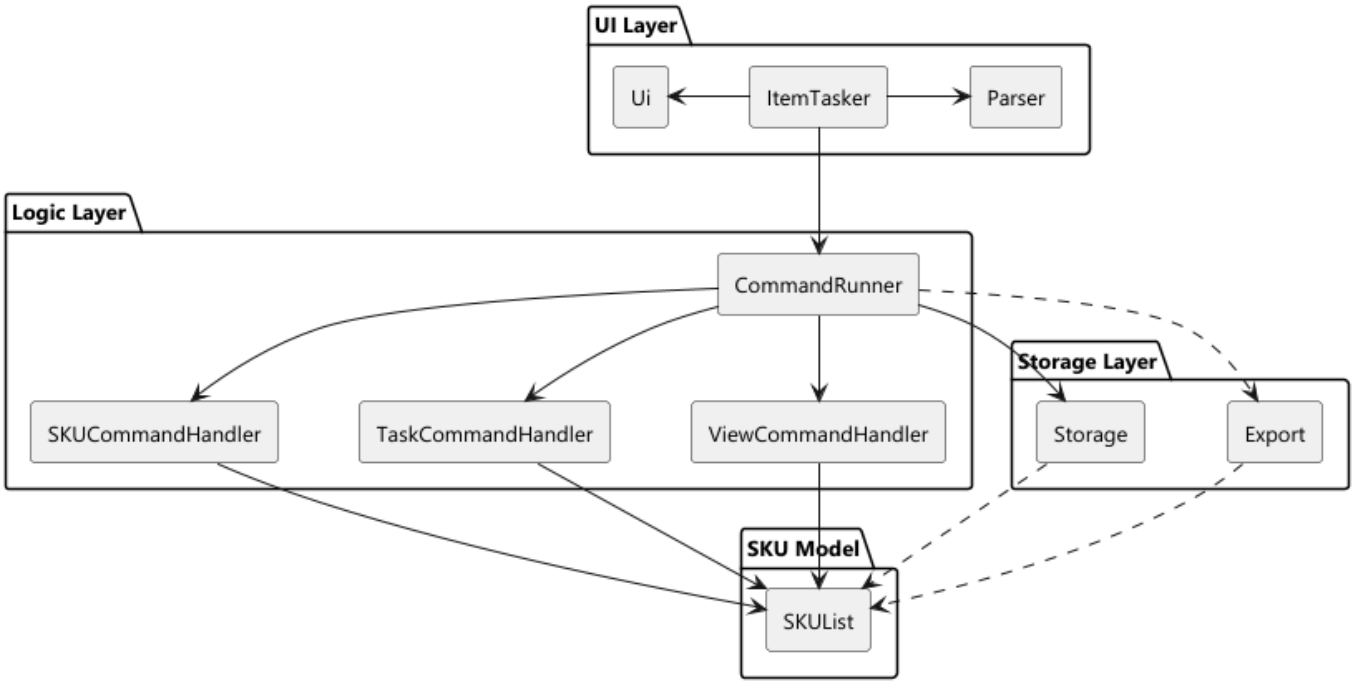
## Command Summary

| Action | Format | Example |
|---|---|---|
| Add SKU | `addsku n/SKU_ID l/LOCATION` | `addsku n/PALLET-A l/B2` |
| Edit SKU | `editsku n/SKU_ID l/NEW_LOCATION` | `editsku n/PALLET-A l/C3` |
| Delete SKU | `deletesku n/SKU_ID` | `deletesku n/PALLET-A` |
| Add task | `addskutask n/SKU_ID d/DUE_DATE [p/PRIORITY] [t/DESCRIPTION]` | `addskutask n/PALLET-A d/2026-06-15 p/HIGH t/Check stock` |
| Edit task | `edittask n/SKU_ID i/TASK_INDEX [d/DUE_DATE] [p/PRIORITY] [t/DESCRIPTION]` | `edittask n/PALLET-A i/1 d/2026-12-31 p/LOW` |
| Delete task | `deletetask n/SKU_ID i/TASK_INDEX` | `deletetask n/PALLET-A i/2` |
| Mark task | `marktask n/SKU_ID i/TASK_INDEX` | `marktask n/PALLET-A i/1` |
| Unmark task | `unmarktask n/SKU_ID i/TASK_INDEX` | `unmarktask n/PALLET-A i/1` |
| Sort tasks | `sorttasks n/SKU_ID s/SORT_FIELD [o/ORDER]` | `sorttasks n/PALLET-A s/priority o/descending` |
| List tasks | `listtasks [n/SKU_ID] [p/PRIORITY] [l/LOCATION]` | `listtasks p/HIGH` |
| Find tasks | `find [n/SKU_ID] [t/DESCRIPTION] [i/TASK_INDEX]` | `find n/PALLET-A t/Check` |
| View status | `status [n/SKU_ID]` | `status n/PALLET-A` |
| View warehouse map | `viewmap` | `viewmap` |
| Export inventory | `export` | `export` |
| Help | `help` | `help` |
| Exit | `bye` / `exit` | `bye` |

# Contributions to the Developer Guide (Extracts)

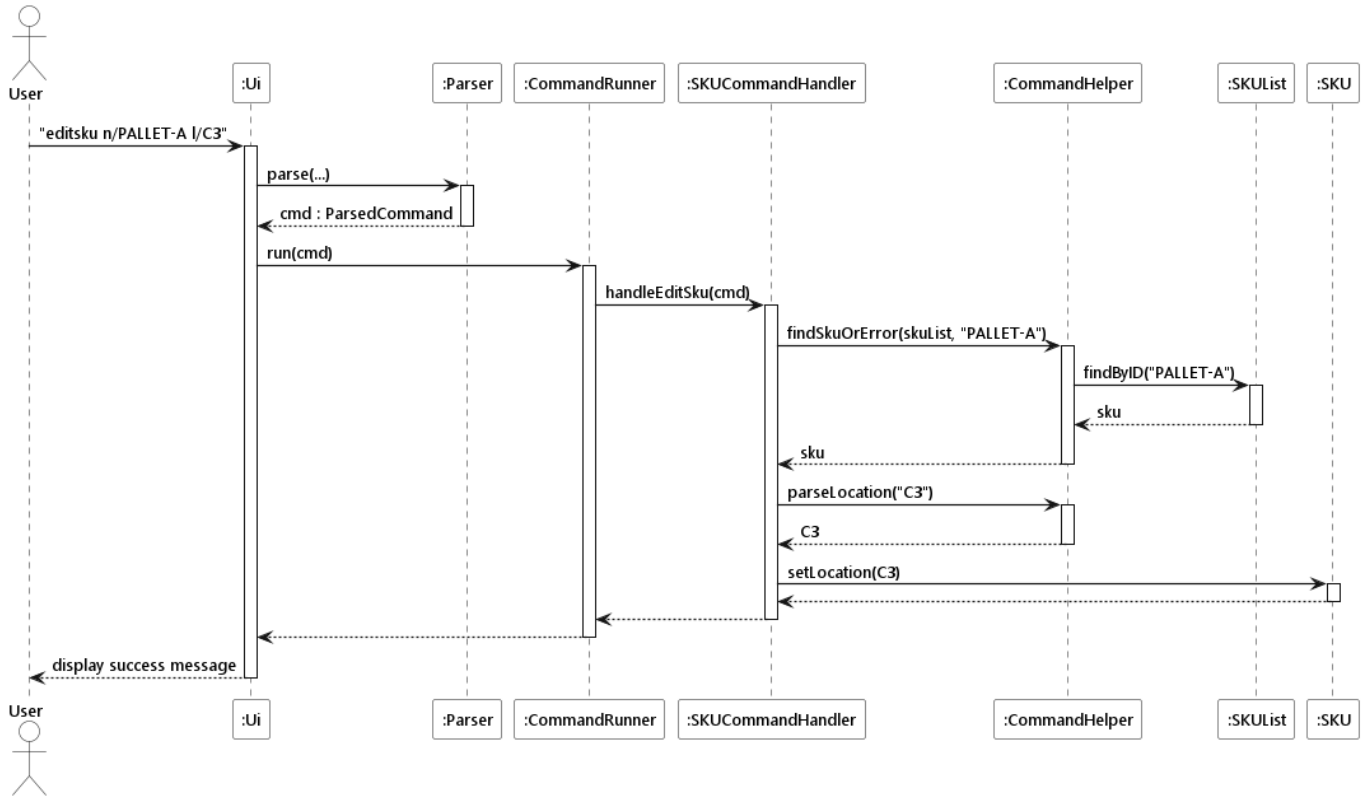
Note that only a small portion is made available here due to page limits.

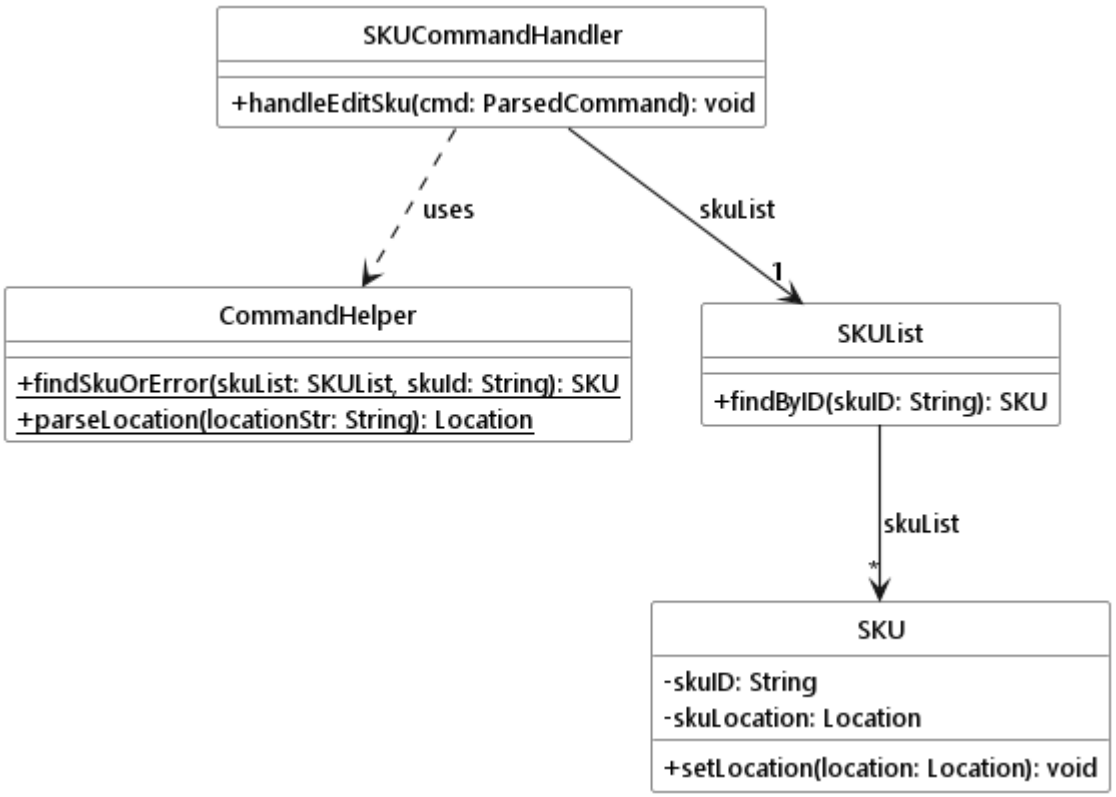
## Design: Architecture



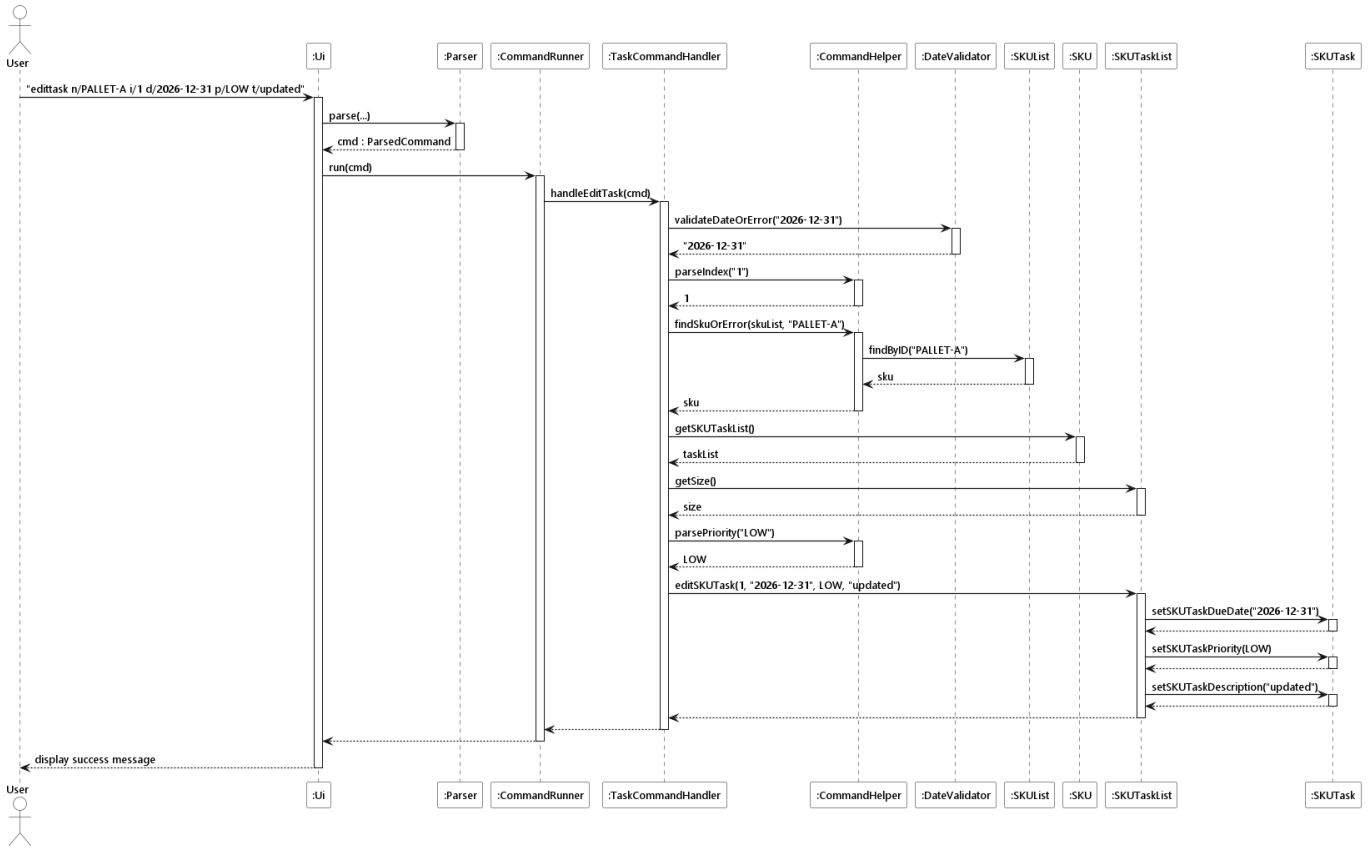
# Edit SKU / Edit Task Feature

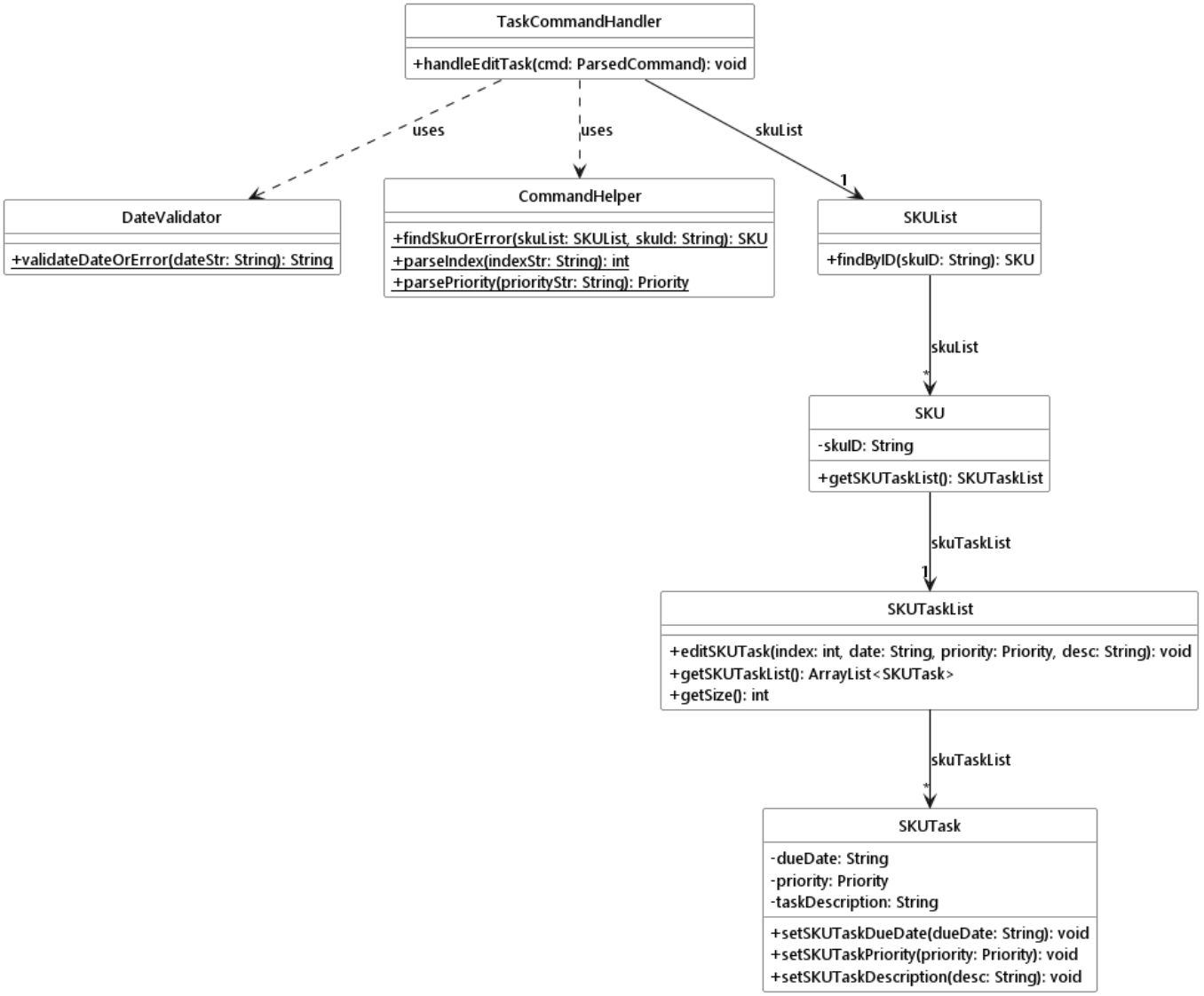
## Edit SKU



## Edit Task


User
:Ui
:Parser
:CommandRunner
:TaskCommandHandler
:CommandHelper
:DateValidator
:SKUList
:SKU
:SKUTaskList
:SKUTask
"edittask n/PALLET-A i/1 d/2026-12-31 p/LOW t/updated"
parse(...)
cmd : ParsedCommand
run(cmd)
handleEditTask(cmd)
validateDateOrError("2026-12-31")
"2026-12-31"
parseIndex("1")
1
findSkuOrError(skuList, "PALLET-A")
findByID("PALLET-A")
sku
sku
getSKUTaskList()
taskList
getSize()
size
parsePriority("LOW")
LOW
editSKUTask(1, "2026-12-31", LOW, "updated")
setSKUTaskDueDate("2026-12-31")
setSKUTaskPriority(LOW)
setSKUTaskDescription("updated")
display success message


TaskCommandHandler
+handleEditTask(cmd: ParsedCommand): void
uses
uses
skuList
1
DateValidator
+validateDateOrError(dateStr: String): String
CommandHelper
+findSkuOrError(skuList: SKUList, skuId: String): SKU
+parseIndex(indexStr: String): int
+parsePriority(priorityStr: String): Priority
SKUList
+findByID(skuID: String): SKU
skuList
*
SKU
-skuID: String
+getSKUTaskList(): SKUTaskList
skuTaskList
1
SKUTaskList
+editSKUTask(index: int, date: String, priority: Priority, desc: String): void
+getSKUTaskList(): ArrayList<SKUTask>
+getSize(): int
skuTaskList
*
SKUTask
-dueDate: String
-priority: Priority
-taskDescription: String
+setSKUTaskDueDate(dueDate: String): void
+setSKUTaskPriority(priority: Priority): void
+setSKUTaskDescription(desc: String): void